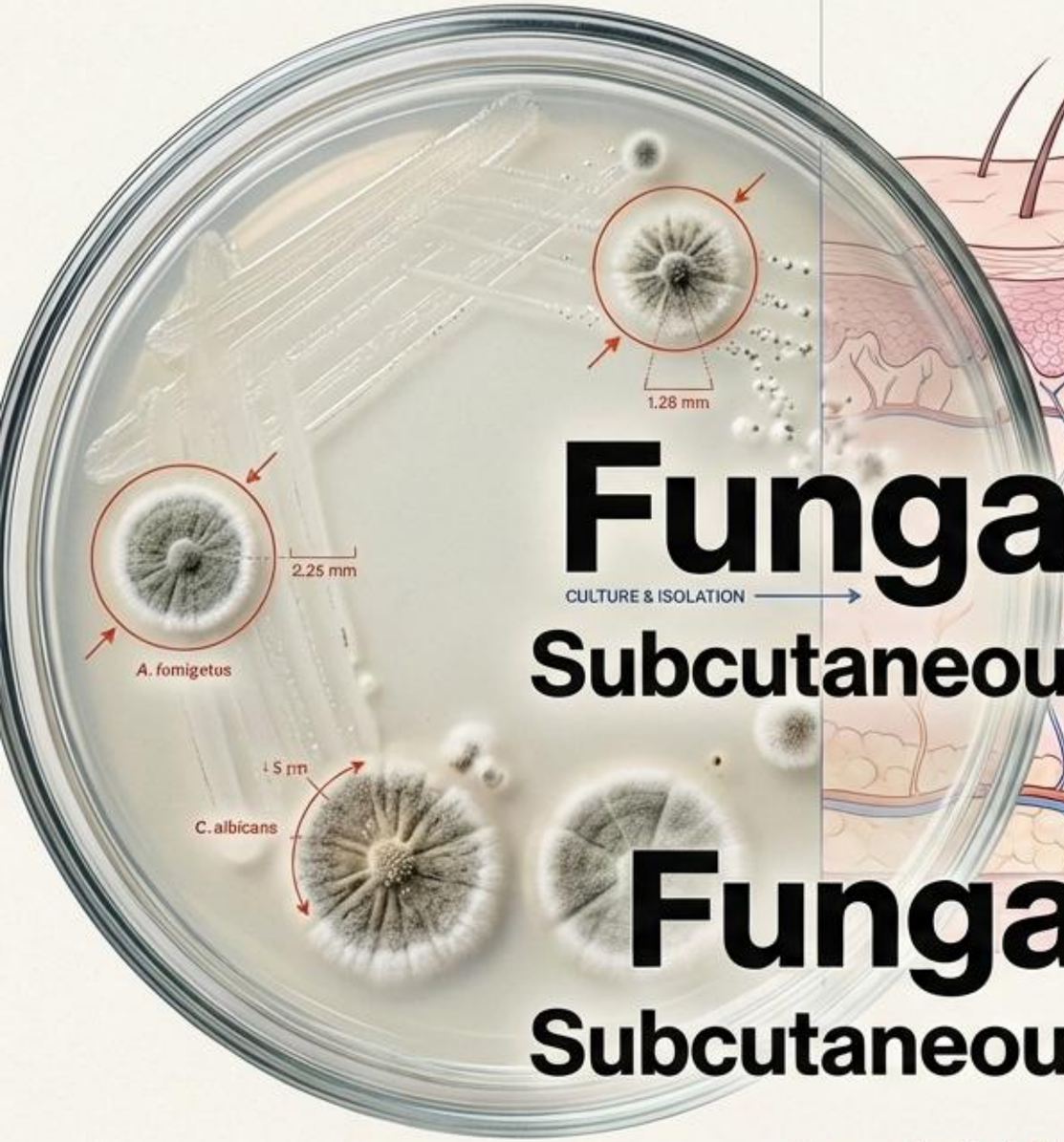
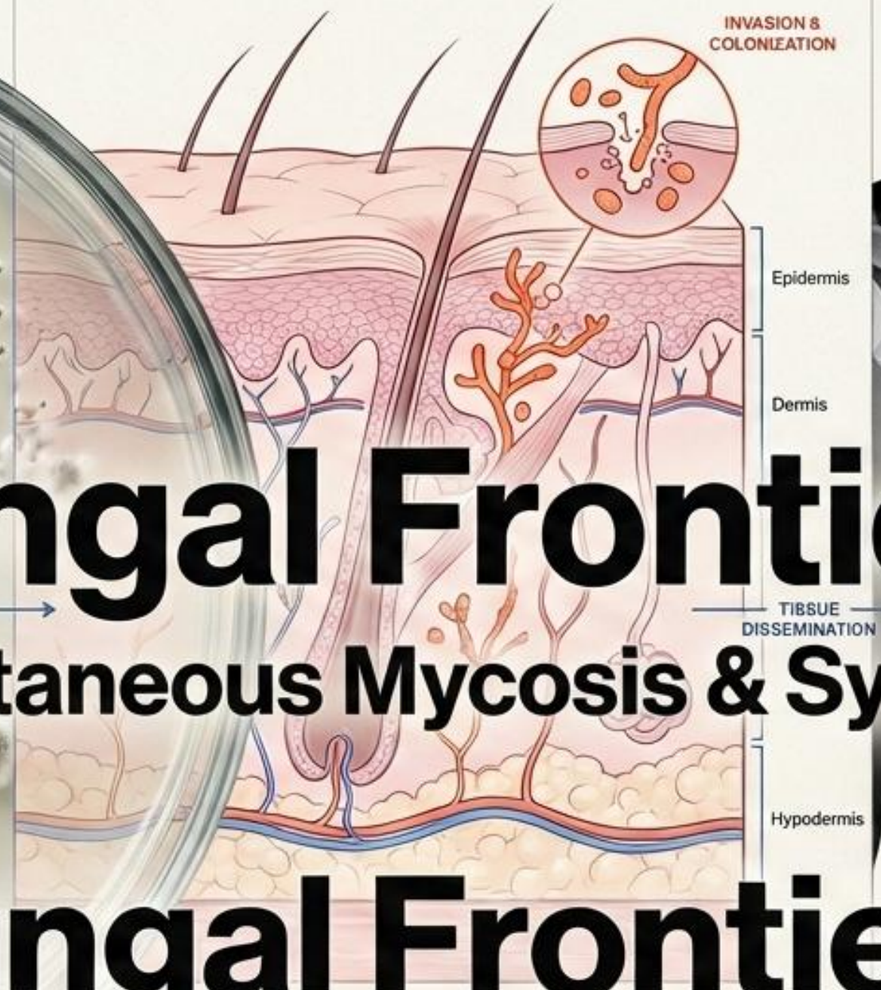
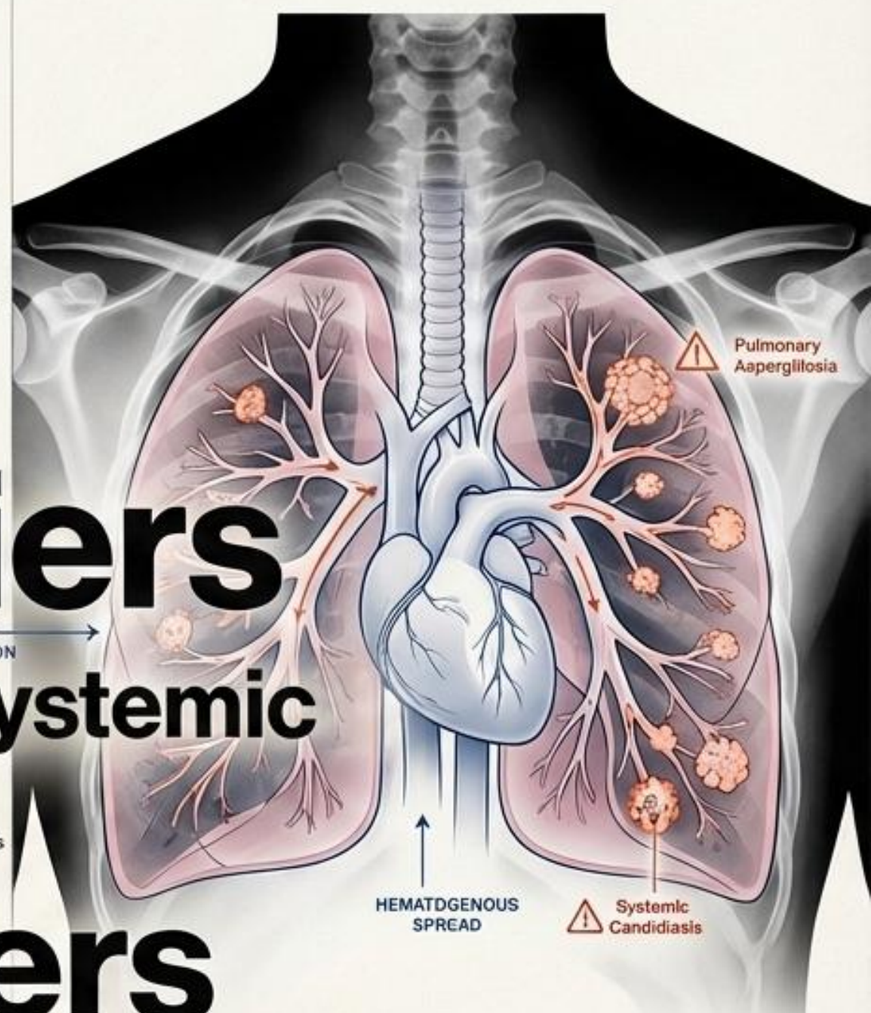






Fungal Frontiers

CULTURE & ISOLATION → TIBSUE DISSEMINATION →

Subcutaneous Mycosis & Systemic

Fungal Frontiers

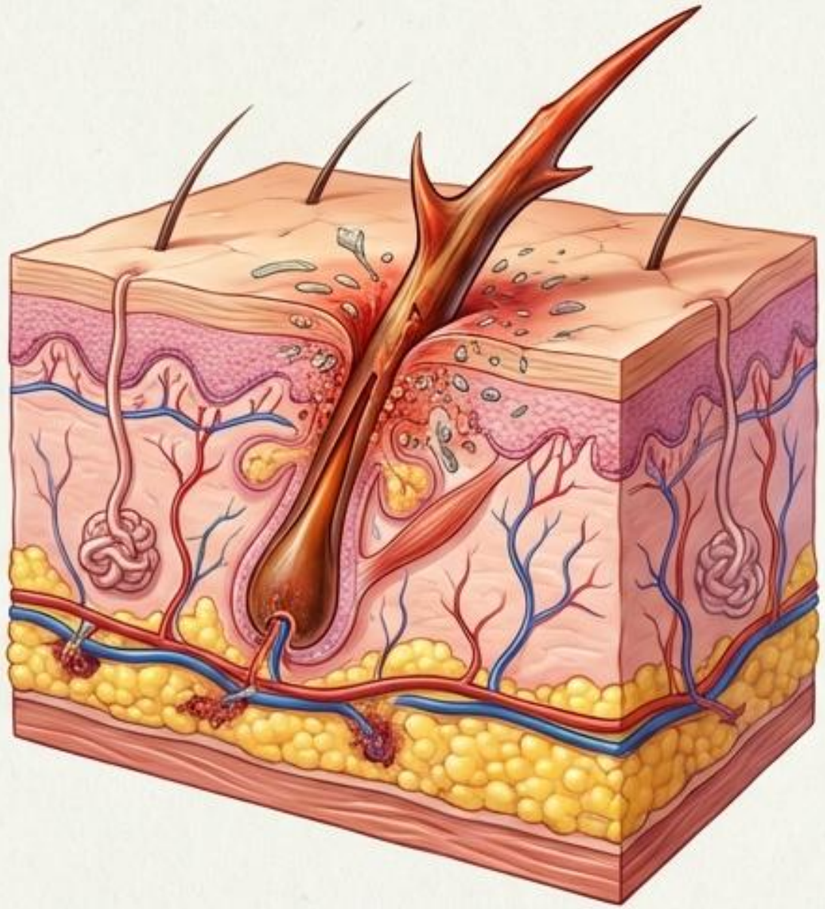
Subcutaneous Mycosis & Systemic

From the lab bench to the systemic circulation.

Subcutaneous Mycoses: The Traumatic Entry

Definition: Infections with characteristic lesions in subcutaneous tissue, overlying skin, and potentially bone/muscle.

Mechanism: Percutaneous introduction via 'Trivial Trauma' (thorns, splinters).



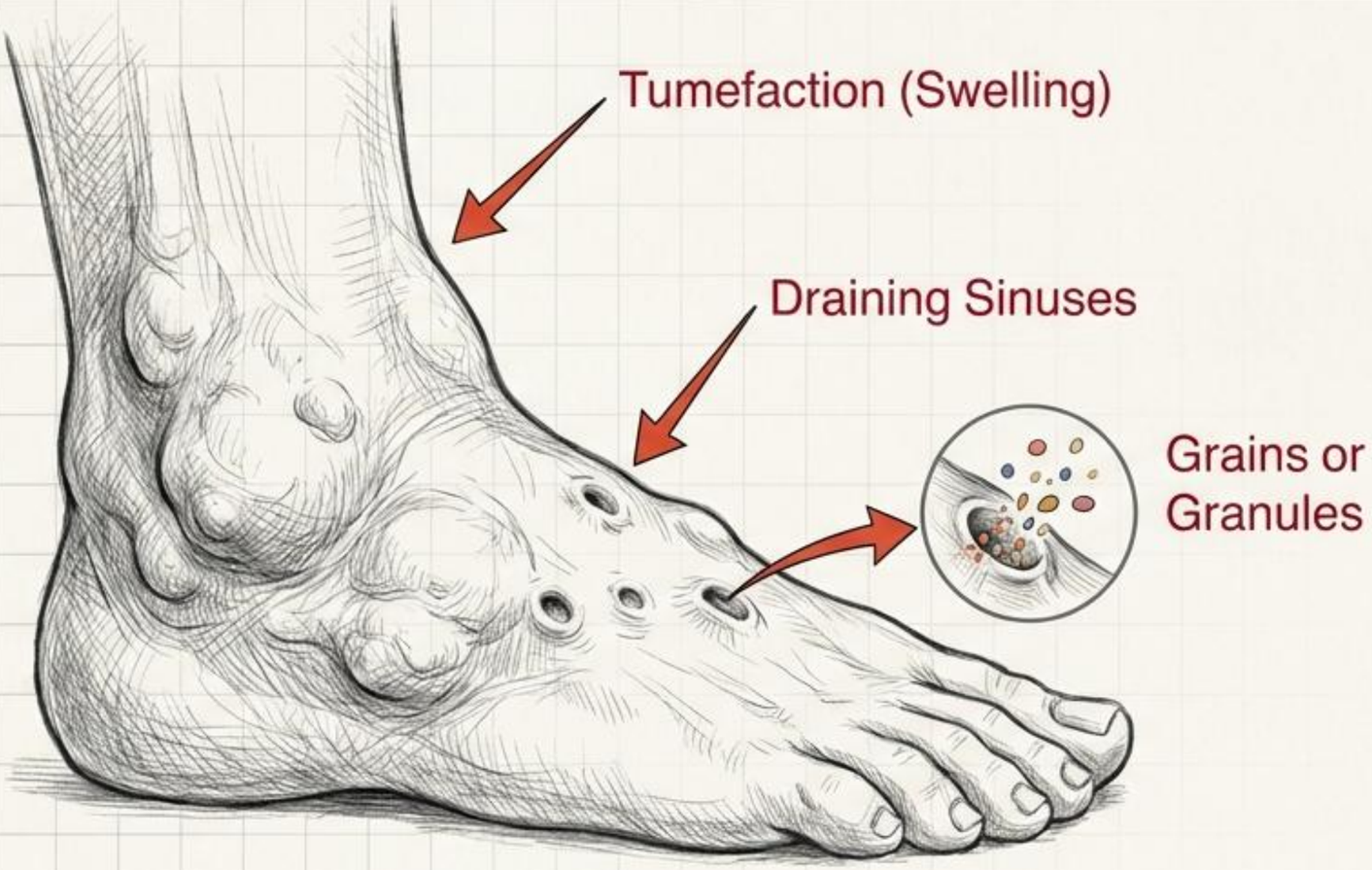
Agents: Heterogeneous group with low pathogenic potential.

The Spectrum of Subcutaneous Agents

Condition	Causative Agents
Mycetoma	Bacteria, fungi
Chromoblastomycosis	*Fonsecaea pedrosoi*, *Fonsecaea compactum*, *Phialophora verrucosa*, *Cladophialophora carrionii*, *Rhinocladiella aquaspersa*
Phaeohyphomycosis	*Exophiala jeanselmei*, *Bipolaris spicifera*, *Wangiella dermatitidis*
Sporotrichosis	*Sporothrix schenckii*
Rhinosporidiosis	*Rhinosporidium seeberi*

Mycetoma: The Clinical Triad

A chronic, progressive granulomatous infection affecting extremities (skin, fascia, bone).



Decoding the Grains: Mycetoma Diagnosis

Black Grain

Madurella spp., *Exophiala*, *Curvularia*

White Grain

Pseudoallescheria, *Acremonium*, *Nocardia* (bacterial)

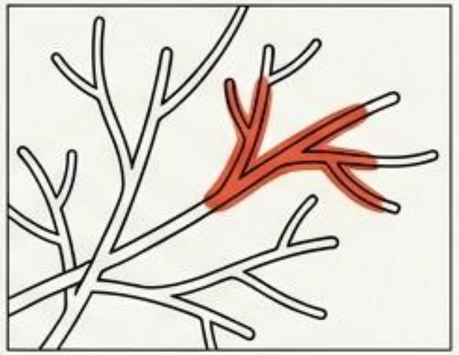
Red Grain

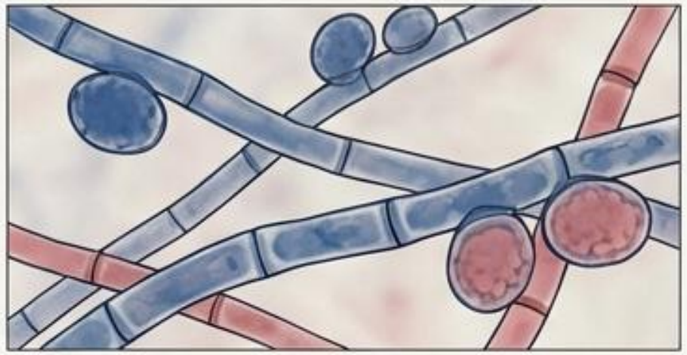
Actinomadura pelletieri

Microscopic Differentiation

Actinomycotic (Bacterial): Thin filaments (<1 µm).

Eumycotic (Fungal): Broad filaments, septate hyphae, chlamydospores.





Sporotrichosis: The Rose Gardener's Disease

Agent:
Sporothrix schenckii

Ecology:
Soil, decaying wood, moss, infected animals.

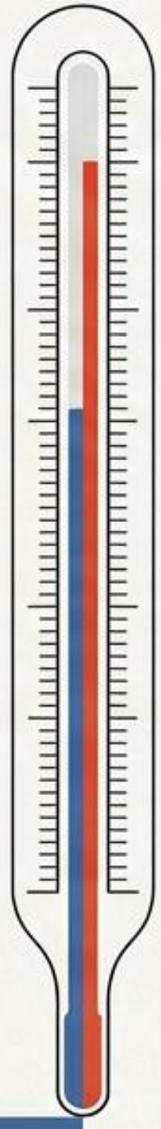
Entry:
Direct skin contact via spores (Prick/Trauma).

Pathology:
Chronic pyogenic granulomatous lesions.

The Shapeshifter: Thermal Dimorphism

Environment / Culture

22–25°C



37°C

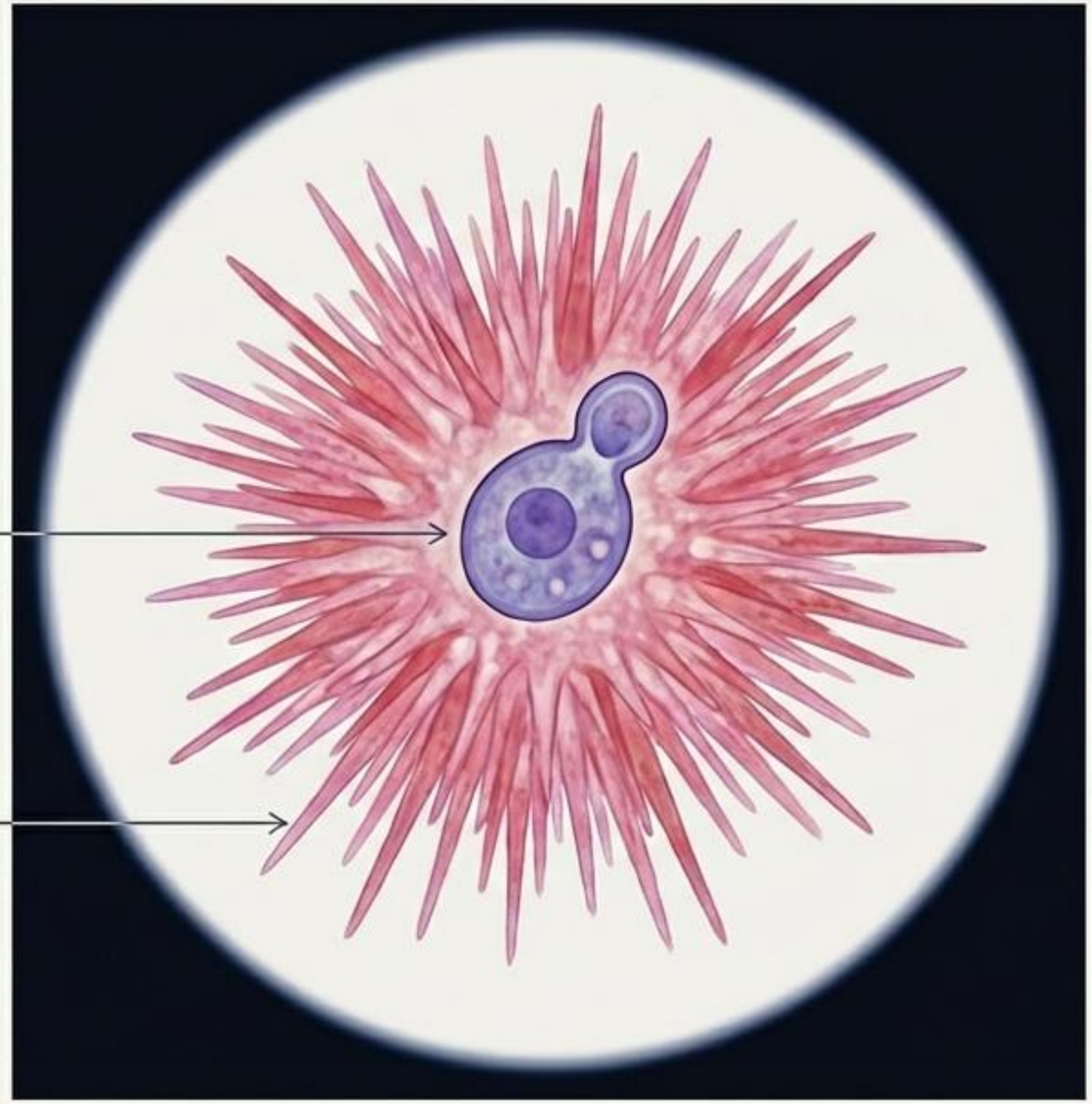
Tissue / Body

Mold Phase
Black/shiny colonies becoming wrinkled.

Yeast Phase
Found in infected tissue.

NotebookLM

The Diagnostic Hallmark: Asteroid Bodies



Crimson Pro

Found in pus from abscesses.

Visualized with PAS or GMS stains.

Systemic Invasion: The Virulent Shift

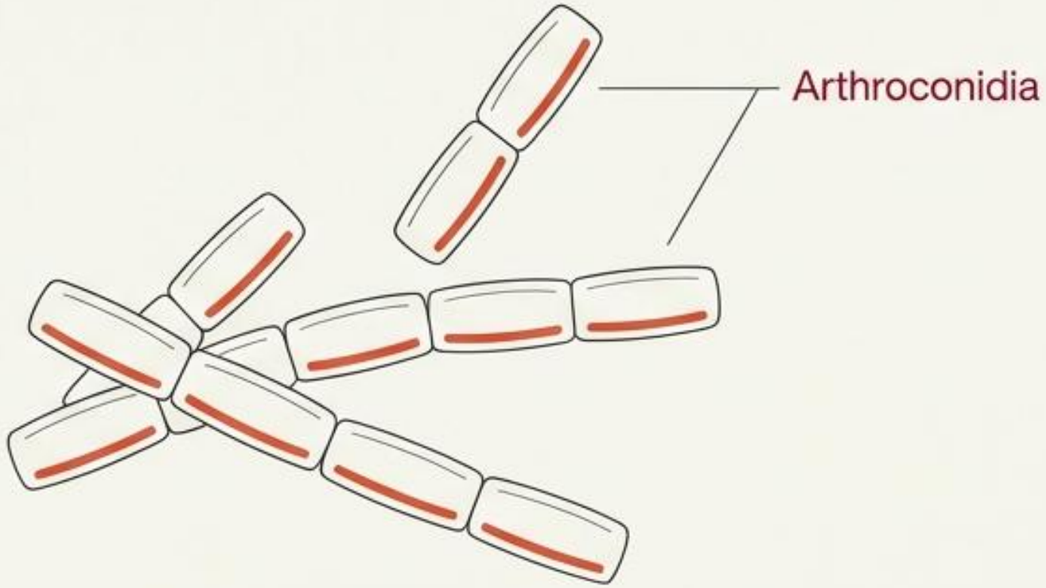
Systemic Mycoses are caused by characteristically virulent soil fungi that can infect HEALTHY humans.

1. **Coccidioidomycosis**
2. **Paracoccidioidomycosis**
3. **Histoplasmosis**
4. **Blastomycosis**
5. **Cryptococcosis**

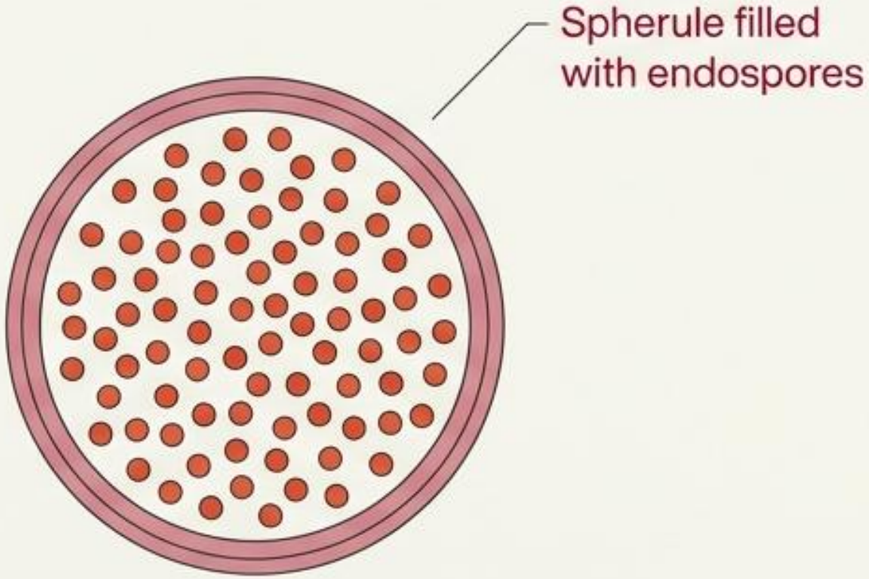
Coccidioidomycosis: The Agent

Coccidioides immitis

Environmental Form (Soil / 25°C)

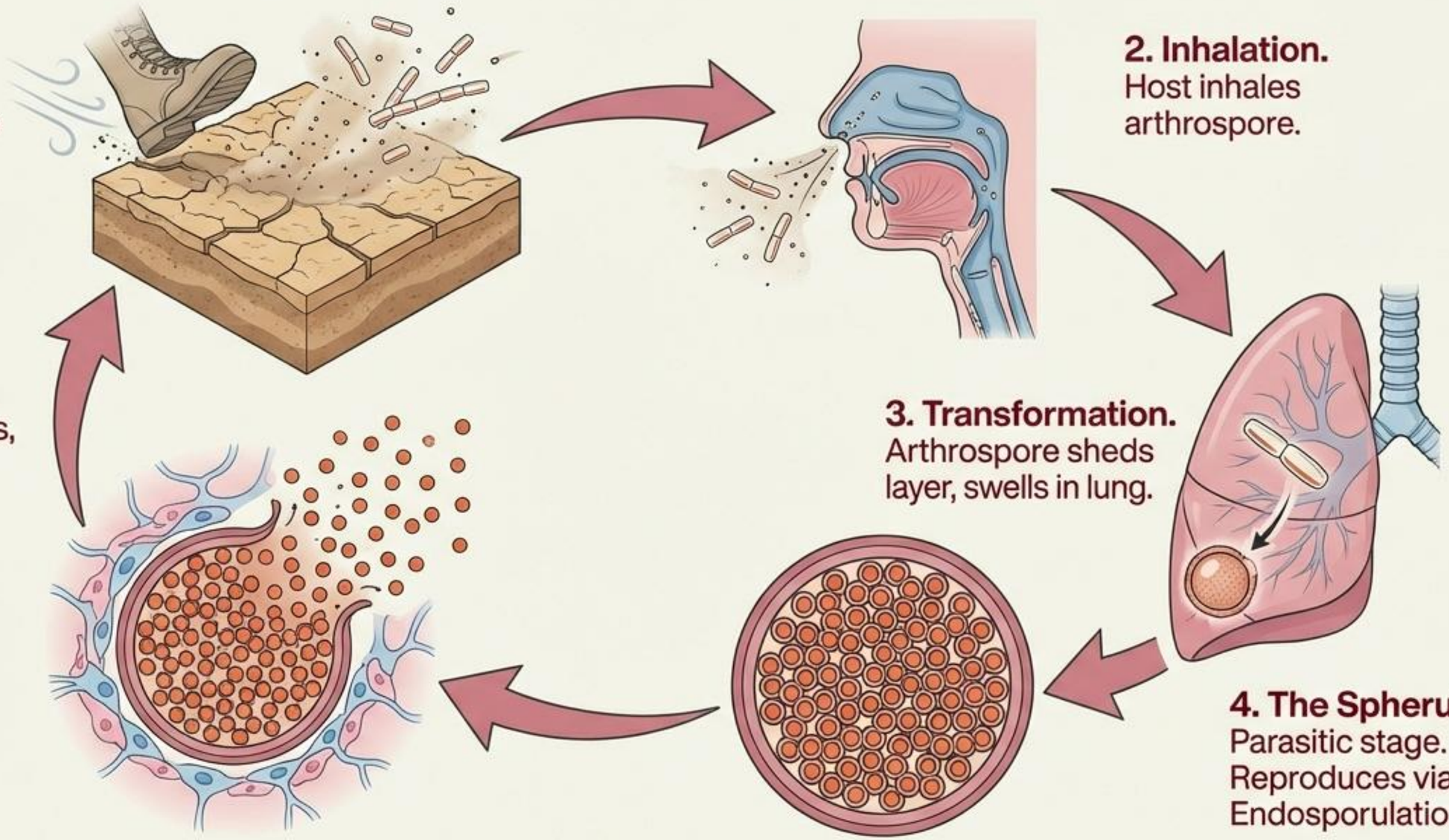


Tissue Form (Body / 37°C)



Dimorphic fungus. Recognized as distinct entity in 1892.

From Soil to Spherule: The Life Cycle



Clinical Impact & Identification

Presentation

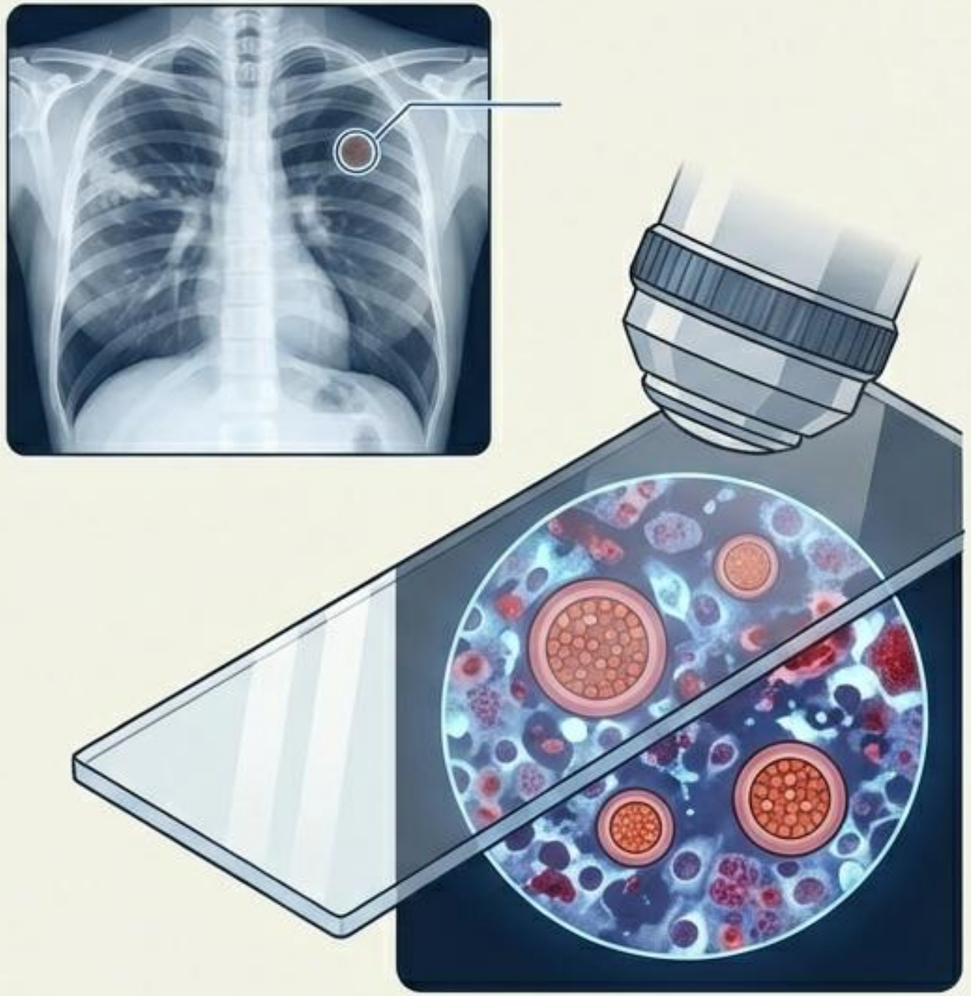
Primary: Pulmonary disease.

Disseminated: CNS, Skin, Bones.

Diagnosis

Microscopy: Spherules in sputum/biopsy (Calcofluor white, H&E, Silver, PAS).

Serology: IgM (Tube-precipitating, EIA) and IgG (Complement fixation).

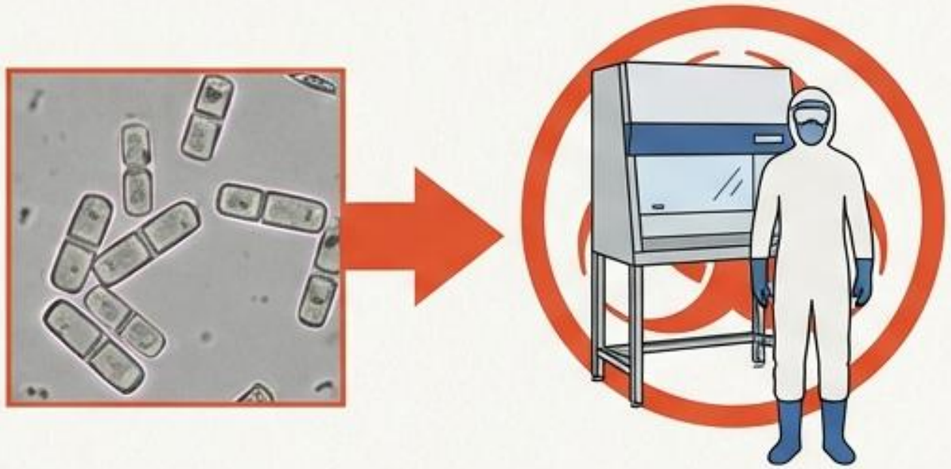


The Double-Edged Sword: Culture Safety

WARNING **WARNING**

Culture is the Gold Standard

(White, cottony colony on SDA in 5 days).



HAZARD WARNING

Culture produces typical Arthroconidia which are HIGHLY INFECTIOUS.

Culturing poses significant risk to laboratory personnel.

Safer Alternative: DNA probes.